

Def. Let R be a Commutative Ring with $1 \neq 0$.

The polynomial ring in the indeterminate x with coefficients from R is:

$$R[x] \stackrel{\text{def}}{=} \bigcup_{n=0}^{\infty} \prod_{k=0}^n R \quad \text{where} \quad x_k^n = \begin{cases} R & \text{if } k \leq n \\ \{0\} & \text{if } k > n. \end{cases}$$

Denote a tuple $(a_0, a_1, \dots, a_n, 0, 0, \dots) \in \prod_{k=0}^{\infty} R$ as:

$$a_0 + a_1x + \dots + a_nx^n$$

And, the addition is given by entry-wise, the multiplication is given by

$$\left(\sum_{i=0}^n a_i x^i \right) \cdot \left(\sum_{i=0}^m b_i x^i \right) = \sum_{k=0}^{n+m} \underbrace{\left(\sum_{i=0}^k a_i b_{k-i} \right)}_{\text{Cauchy Product}} \cdot x^k$$

Prop. Let $I \subset R$ be an Ideal of R . Then, in $R[x]$,

$$(I) = I[x]$$

Moreover $R[x]/(I) = R[x]/I[x] \cong (R/I)[x]$.

Proof. $(I) = I[R[x]] = (IR)[x] = I[x]$, since R is commutative, and has 1.

Now, define a map $\varphi: R[x] \rightarrow (R/I)[x]: \sum a_n x^n \mapsto \sum (a_n + I)x^n$

Then, $\ker \varphi = I[x]$, so proved by the first isomorphism theorem.

(More specifically,

$$\sum a_n x^n \in \ker \varphi \Leftrightarrow \sum (a_n + I)x^n = 0$$

$$\Leftrightarrow \text{all coefficients } a_n \text{ are contained in } I$$

$$\Leftrightarrow \sum a_n x^n \in I[x].$$



Thm. If R is Noetherian Ring, then $R[x]$ is Noetherian.

Proof Let an ideal $I \subseteq R[x]$ be given, and let $L \subseteq R$ be the set of all leading coefficients of the elements in I . (Claim L is ideal of R).

Since $0 \in I$, so $0 \in L$. Let $a, b \in L$ and $r \in R$ be given.

Then, there are two polynomials in I ,

$$f(x) = ax^d + \dots \quad \text{and} \quad g(x) = bx^e + \dots$$

Since I is Ideal,

$$r \cdot x^e \cdot f(x) - x^d \cdot g(x) \text{ is contained in } I, \text{ so } ra - b \in L.$$

Which shows $L \subseteq R$ is an Ideal. Now, L is finitely generated by assumption for R .

Denote $L = (a_1, \dots, a_n)$ where $a_1, \dots, a_n \in R$.

And, pick $f_i \in I$ be a polynomial of degree e_i with the leading coefficient is a_i .

Let $N = \max\{e_1, \dots, e_n\}$. For each $d \in \{0, 1, \dots, N-1\}$, construct

L_d be the set of all leading coefficients of polynomial in I of degree d , with 0 .

Similarly, each L_d is Ideal of R , so these are generated finitely:

$$L_d = (b_{d,1}, \dots, b_{d,n_d}) \text{ for some } b_{d,i} \in R.$$

Now, for $f_{d,i}$, the polynomial of degree d with the leading coefficient is $b_{d,i}$, the set $\{f_1, \dots, f_n\} \cup \{f_{d,i} \mid 0 \leq d < N, 1 \leq i \leq n_d\}$ generates I , because:

Clearly the generated ideal is contained in I . If there is a polynomial $f(x) \in I$ which is not contained in the generated ideal with the minimal degree.

If $\deg f \geq N$, then the leading coefficient a of f can be represent as:

$$a = r_1 a_1 + \dots + r_n a_n \quad (r_i \in R), \text{ } R\text{-linear combinations of generators for } I.$$

Then, $g = r_1 x^{d-e_1} f_1 + \dots + r_n x^{d-e_n} f_n$ has a degree smaller than $\deg f$, with the leading coefficient is a . Contradiction to the minimality.

If $\deg f < N$, $a \in L_d$ ($d < N$), so contradiction.

□